

Language Exchange Platform – Project Proposal

Project Overview

We are building a language exchange platform where people from around the world can connect with each other to practice speaking different languages. The primary goal is to help users improve their communication skills through real conversations with native or fluent speakers.

Unlike traditional learning apps, this platform focuses on peer-to-peer interaction, making language learning more engaging and practical.

The long-term vision is to build a platform similar to PeerUp or Tandem, with AI-powered learning assistance.

Problem Statement

Many people struggle to improve their speaking skills because they don't have someone to practice with regularly. Existing platforms are either paid, limited, or lack intelligent learning features.

Our platform will solve this by:

- Connecting learners with suitable language partners.
 - Providing text, voice, and video communication.
 - Using AI to help users improve their grammar, vocabulary, and fluency.
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Target Users

- Students learning a new language.
 - Professionals preparing for interviews.
 - Travelers.
 - Anyone interested in language exchange.
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Project Goals

- Build a scalable language learning platform.
- Provide a smooth user experience.
- Implement real-time communication.
- Integrate AI features in future releases.

- Deploy the application for public use.
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Core Features (MVP)

Authentication

- User Registration
- Login
- JWT Authentication
- Forgot Password
- Email Verification

User Profile

- Name
- Profile Picture
- Bio
- Native Language
- Languages Learning
- Proficiency Level
- Interests

Language Matching

- Search users
- Filter by language
- Filter by country
- Suggested language partners

Friend System

- Send request
- Accept/Reject request
- Friends list

Real-Time Chat

- One-to-one messaging
- Online/Offline status
- Typing indicator
- Read receipts

Practice Sessions

- Schedule speaking sessions
- Session reminders
- Session history

Future Features

- Voice Calls
- Video Calls
- Group Conversations
- Community Rooms
- Language Challenges
- Leaderboards
- Streak System
- Badges & Rewards
- Notifications
- Admin Dashboard

AI Features (Phase 2)

- AI Grammar Correction
- Vocabulary Suggestions
- Pronunciation Feedback
- Conversation Summary
- Speaking Score
- AI Language Tutor
- Personalized Learning Recommendations
- Daily Practice Topics

Technology Stack

Backend

- Python
- FastAPI

Frontend

- React

Database

- PostgreSQL

Cache

- Redis

Authentication

- JWT

Real-Time Communication

- WebSockets

Voice & Video

- WebRTC

Containerization

- Docker

Version Control

- Git & GitHub

Project Management

- JIRA

CI/CD

- GitHub Actions

API Testing

- Postman

Proposed Architecture

Frontend (React)

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FastAPI Backend

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PostgreSQL

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Redis

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WebSockets

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WebRTC

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AI Services (Future)

Team Responsibilities

Developer 1 (Backend)

- Database Design
- Authentication
- User Management
- REST APIs
- Matching System
- Redis Integration
- Docker
- CI/CD
- Deployment

Developer 2 (Frontend)

- React UI
- Authentication Pages
- Dashboard
- Chat Interface
- Voice/Video UI
- Responsive Design
- API Integration

Both developers will:

- Review code
 - Fix bugs
 - Write documentation
 - Test features
 - Deploy updates
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Development Roadmap

Phase 1 (Weeks 1–2)

- *Project setup*
- Database schema
- Authentication
- User profile
- JWT

Phase 2 (Weeks 3–4)

- Language matching
- Friend system
- Search & filters
- REST APIs

Phase 3 (Weeks 5–6)

- Real-time chat
- Online status
- Typing indicator
- Notifications

Deliverable: MVP ready for testing.

Phase 4 (Weeks 7–9)

- Voice & video calls
- Practice session scheduling
- UI improvements
- Deployment

Phase 5 (Future)

- AI-powered learning features

- Performance optimization
 - Scaling and production readiness
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Development Practices

- Feature-based Git branching.
 - Pull requests for all major changes.
 - Code reviews before merging.
 - Meaningful commit messages.
 - Environment variables for secrets.
 - API documentation using Swagger/OpenAPI.
 - Regular testing before deployment.
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Success Criteria

The project will be considered successful when users can:

- Create an account.
 - Build a language-learning profile.
 - Find compatible language partners.
 - Send friend requests.
 - Chat in real time.
 - Schedule practice sessions.
 - Use the platform smoothly on desktop and mobile.
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Future Vision

Our long-term goal is to evolve this platform into an AI-powered language learning ecosystem that combines peer-to-peer communication with intelligent feedback, personalized learning, and scalable backend architecture. This project will also serve as a production-quality portfolio application demonstrating backend development, real-time systems, DevOps, and applied AI.